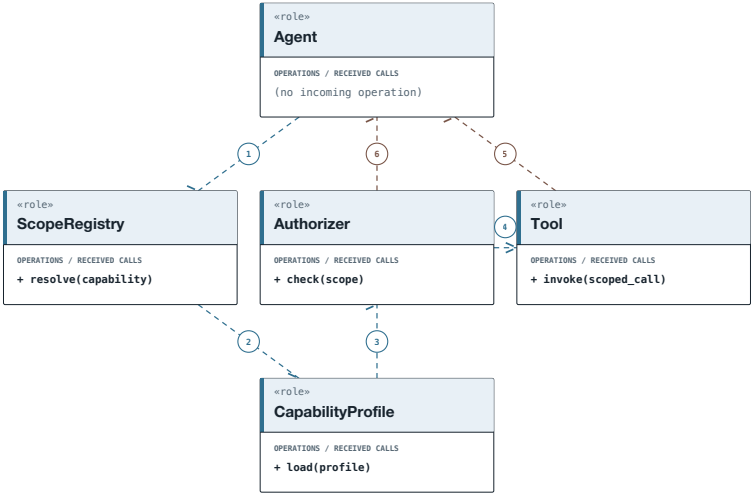
P25 Scope-based authorization and profiles

Primitive / UML CLASS VIEW / ANALYTICAL MODEL

Bind capabilities to explicit scopes.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A caller invokes a method or service with an operator, session, or agent identity.
TRIGGER	Capability is checked against a requested scope.
STRUCTURE	Scope identity, profile, grant and revocation state, and resolution function.
EFFECT	The call is permitted only if the principal currently holds a covering scope.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Principal identity and requested capability are explicit.
ENSURES	Authorization is evaluated at the call boundary and records the covering scope.
INVARIANT	Scopes are not inherited beyond declared containment or lifetime.
VIOLATION	Missing, expired, revoked, or cross-scope capabilities are denied explicitly.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 Agent → ScopeRegistry resolve(capability)
2 ScopeRegistry → CapabilityProfile load(profile)
3 CapabilityProfile → Authorizer check(scope)
4 Authorizer → Tool invoke(scoped_call)
5 Tool ↩ Agent return(result)
6 Authorizer ↩ Agent deny(failure)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Method scopes, roles, session grants, profiles, and delegated identities.
TEST_OBLIGATION	Test widening, revocation, expiration, child agent, and cross-tenant access.
NEGATIVE_CONTROL	A boolean admin flag is not a scope model.